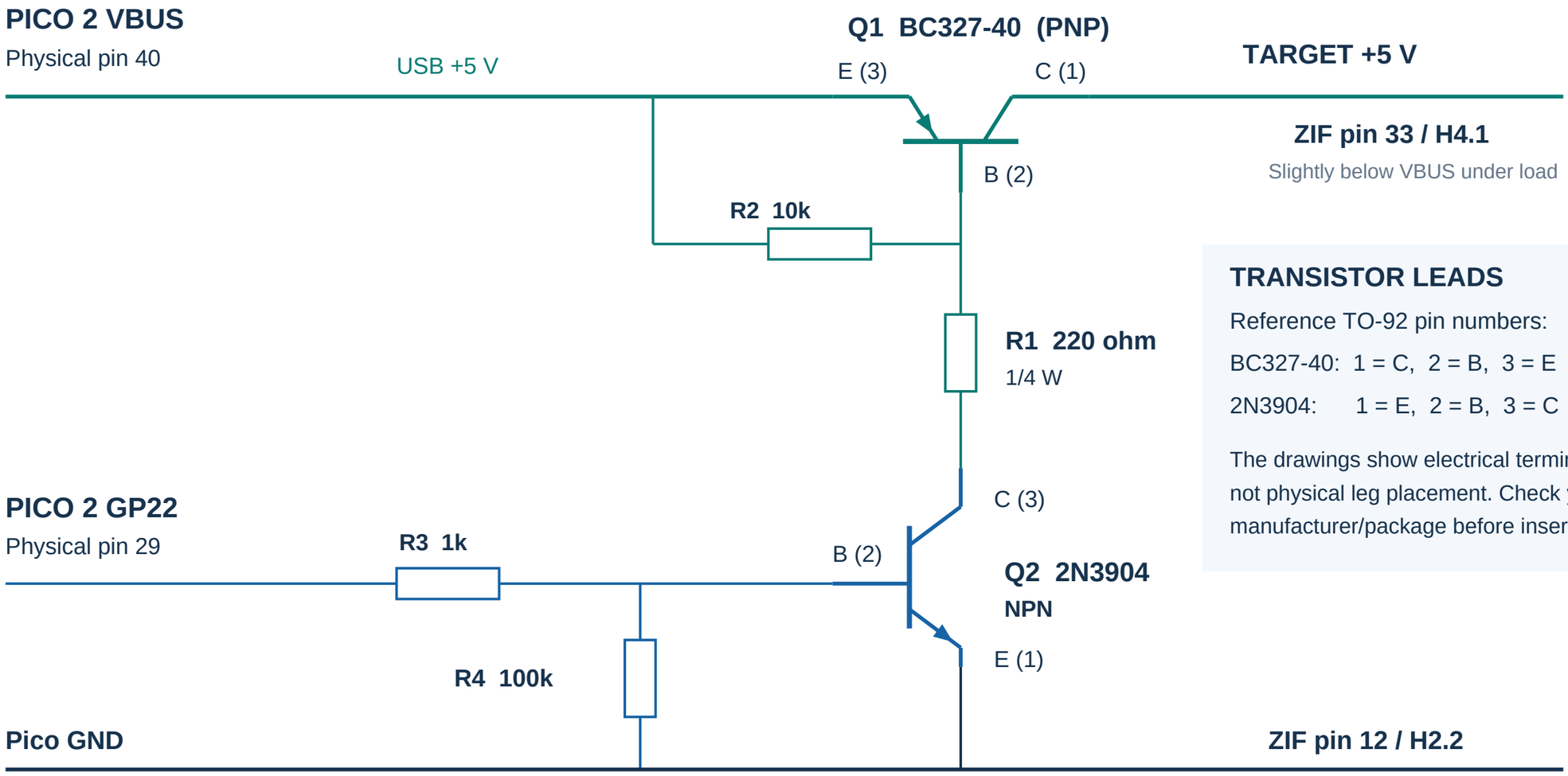
OPTIONAL TARGET POWER / PARTS ALREADY ON HAND

BC327-40 + 2N3904 + four resistors | GP22 HIGH = power ON | LOW / floating = OFF

Replace the page-1 power jumper with this switch. Leave target USB unplugged; retain all signal/ground wires.
Q1 switches +5 V. Ground remains permanently connected. No target 3.3 V connection.



TRANSISTOR LEADS

Reference TO-92 pin numbers:

BC327-40: 1 = C, 2 = B, 3 = E

2N3904: 1 = E, 2 = B, 3 = C

The drawings show electrical terminals, not physical leg placement. Check your manufacturer/package before inserting.

Common ground: same as page 1

HOW IT WORKS

GP22 drives Q2, which pulls Q1 base down through R1. Q1 then supplies the target. R2 and R4 default both transistors off.
R1 gives about 18 mA base drive to Q1; R3 draws about 2.5 mA from the Pico. Initial design load: up to 150 mA target current.

Check the loaded voltage at ZIF pin 33 before relying on this switch: aim for at least 4.5 V with the target running.

Before GP22 LOW, release bus/SWD outputs and disable pulls. Start with a 1-second off delay; verify that the target resets.

Pico USB must supply probe + target + base drive; request a 500 mA USB configuration before enabling target power.

Prototype switch: no current limiting or active discharge. No relay or flyback diode is needed. See README for connections.